

PHYS 357: Honours Quantum Physics 1

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Abstract

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1 Introduction

Textbook used is JS Townsend, *A Modern Approach to Quantum Mechanics*, 2nd ed., University Science Books 2012. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on Wednesdays, best 8 of 12).
- 20% midterms (best of 2, October 2nd and November 6th).
- 30% final exam.

2 Mathematical Background

2.1 Taylor series

Definition 1 (Taylor series). *A function f that is smooth near a (it can be differentiated any number of times there) equals the power series*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n, \quad (1)$$

where $f^{(n)}$ is the n th derivative. The special case $a = 0$ is called the Maclaurin series.

Result 1 (Series used in this course).

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad (2)$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x| < 1), \quad (1+u)^N = \sum_{n=0}^N \binom{N}{n} u^n \quad (N \in \mathbb{N}). \quad (3)$$

Result 2 (Euler's formula). *Comparing the series for $e^{i\theta}$, $\cos \theta$ and $\sin \theta$ gives $e^{i\theta} = \cos \theta + i \sin \theta$, hence*

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (4)$$

These are used constantly when rotation matrices are written out.

Problem 1 (The exponential as a limit). *Show that $\lim_{N \rightarrow \infty} (1 + x/N)^N = e^x$.*

Remark 1. *This limit is what lets a rotation be written as an exponential (Section 6.2).*

2.2 Bases

Definition 2 (Basis). *A set of vectors is linearly independent if none of them can be written as a combination of the others, and it spans a space if every vector in the space is some combination of them. A basis is a set that does both, so every vector is a unique combination of the basis vectors. The number of basis vectors is the dimension of the space.*

Definition 3 (Kronecker delta).

$$\delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases} \quad (5)$$

3 Stern-Gerlach Experiments

3.1 The classical picture

3.1.1 Angular momentum

Definition 4 (Orbital angular momentum). *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \quad (6)$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x - y plane, only the z component survives. On a circular path of radius r with speed v ,

$$L_z = mvr.$$

Definition 5 (Angular momentum of a rigid body). *For a rigid body¹ rotating about a fixed axis,*

$$L = I_0\omega,$$

with ω the angular velocity and I_0 the moment of inertia, which measures how the mass is distributed relative to the rotation axis.

3.1.2 Magnetic moment of a circulating charge

Definition 6 (Magnetic moment). *A vector $\vec{\mu}$ measuring the net circulation of electric charge. It sets the torque, and in a nonuniform field the force, that a magnet experiences in an external magnetic field.*

A point particle of mass m and charge q on a circular orbit of radius r and period T passes any given point once per period, so it acts as a steady current $I = q/T = qv/2\pi r$. Its moment is

$$\mu = \frac{IA}{c} = \frac{1}{c} \left(\frac{qv}{2\pi r} \right) (\pi r^2) = \frac{q}{2mc} mvr = \frac{q}{2mc} L_z,$$

where A is the area enclosed by the orbit.

¹A rigid body is one that does not deform under the action of external forces.

²Gaussian (CGS) units, where a current loop has moment IA/c . In SI units it is IA .

3.1.3 Gyromagnetic ratio

Definition 7 (Gyromagnetic ratio). *The constant γ relating magnetic moment to angular momentum,*

$$\vec{\mu} = \gamma \vec{L}, \quad \gamma = \frac{q}{2mc} \text{ for a circulating charge.} \quad (7)$$

Remark 2. *This is purely classical: a point charge on a definite orbit, nothing quantized. It matters because measuring $\vec{\mu}$ is how we measure angular momentum, which is what the Stern-Gerlach apparatus does.*

3.2 Spin

Definition 8 (Spin). *Spin $\vec{S}$ is an intrinsic angular momentum: a fixed property of the particle type, like charge or mass, not caused by any motion through space. A stationary electron alone in empty space still has spin. It comes in addition to any orbital $\vec{L}$, not in place of it.*

3.2.1 Intrinsic moment and the g-factor

Spin also carries a magnetic moment proportional to it, but with an extra dimensionless factor,

$$\vec{\mu} = \gamma \vec{S}, \quad \gamma = \frac{gq}{2mc}. \quad (8)$$

Definition 9 (g-factor). *The dimensionless number g above. It says how much stronger or weaker the moment is than the classical value ($g = 1$). It must be measured or derived from relativistic quantum theory.*

3.2.2 Magnetons

Definition 10 (Planck constant). *The constant that sets the scale at which quantization matters. It relates photon energy to frequency, $E = h\nu$, and*

$$h \approx 6.626 \times 10^{-34} \text{ J s}, \quad \hbar = \frac{h}{2\pi} = 1.055 \times 10^{-27} \text{ erg s} = 6.582 \times 10^{-16} \text{ eV s}.$$

Both have units of angular momentum, $[\hbar] = [L_z] = \text{kg m}^2/\text{s}$.

Writing $\vec{S} = \hbar(\vec{S}/\hbar)$ with $\vec{S}/\hbar$ dimensionless, the $\hbar$ joins the gyromagnetic ratio. The combination $|e|\hbar/2mc$ then carries all the units and sets the natural size of the moment for a given mass.

Definition 11 (Magnetons). *The natural units of magnetic moment for the electron (Bohr magneton) and for nucleons, i.e. protons and neutrons (nuclear magneton):*

$$\mu_B = \frac{|e|\hbar}{2m_e c}, \quad \mu_N = \frac{|e|\hbar}{2m_p c} \approx \frac{\mu_B}{1836}, \quad (9)$$

since $m_p/m_e \approx 1836$.

The electron moment is then

$$\vec{\mu}_e = g_e \mu_B \frac{\vec{S}}{\hbar}, \quad (10)$$

and similarly for the proton and neutron with μ_N in place of μ_B .

Particle	Charge q	Mass m	g -factor	Gyromagnetic ratio γ
Electron	$- e $	m_e	$g_e \approx 2$	$g_e \mu_B / \hbar$
Proton	$+ e $	m_p	$g_p \approx 5.59$	$g_p \mu_N / \hbar$
Neutron	0	$m_n \approx m_p$	$g_n \approx -3.83$	$g_n \mu_N / \hbar$

Table 1: Charges, masses, g -factors and gyromagnetic ratios.

Remark 3. Classically $\mu = (q/2mc)L_z$ vanishes when $q = 0$, yet the neutral neutron has a moment. So its $\vec{\mu}$ cannot come from a circulating point charge: a first sign the classical picture fails.

Remark 4. Since $m_p \gg m_e$, $\mu_N \ll \mu_B$, so nuclei are deflected far less than electrons in a Stern-Gerlach device.

3.3 The Stern-Gerlach apparatus

Definition 12 (Stern-Gerlach (SG) apparatus). A device that measures one component of a particle's magnetic moment. A beam of neutral atoms passes through a magnet shaped so that the field strength changes along a chosen axis, here z (a field gradient). The gradient pushes each atom by an amount proportional to μ_z , and a screen records where it lands. Since $\vec{\mu} = \gamma \vec{S}$, the landing position measures S_z .

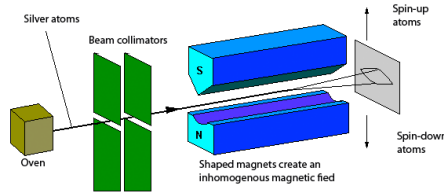


Figure 1: Illustration of the Stern-Gerlach apparatus.

3.3.1 Force in a nonuniform field

A magnetic moment in a field has potential energy $U = -\vec{\mu} \cdot \vec{B}$, so

$$\vec{F} = -\vec{\nabla}U = \vec{\nabla}(\vec{\mu} \cdot \vec{B}), \quad F_z = \mu_z \frac{\partial B_z}{\partial z}.$$

Remark 5. A uniform field gives no net force, only a torque (a twist that turns $\vec{\mu}$ without moving the atom). That is why the magnet needs a gradient.

Definition 13 (Deflection). *The sideways displacement of an atom on the screen caused by F_z . It is proportional to μ_z , so the screen pattern is a record of the moments in the beam.*

3.3.2 Silver atoms

Silver has zero net orbital angular momentum, $\vec{L} = 0$, and its whole moment comes from one unpaired outermost (valence) electron, so the atom behaves like a single electron spin. With

$$\mu_z = -\frac{g_e |e|}{2m_e c} S_z$$

and a gradient $\partial B_z / \partial z < 0$,

$$F_z = g_e \frac{|e|}{2m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z.$$

Result 3. $F_z \propto S_z$: the deflection measures the z component of spin directly.

3.3.3 Classical prediction versus observation

Atoms leave the oven with spins pointing in random directions, so $S_z = |\vec{S}| \cos \theta$ takes every value in $[-|\vec{S}|, |\vec{S}|]$. Classically the screen should show a continuous smear. It shows two spots.

Result 4 (Quantization of spin). *The electron's spin component is quantized:*

$$S_z = \pm \frac{\hbar}{2}. \quad (11)$$

Definition 14 (Spin quantum number). *The number s , fixed by the particle type, taking values $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$. A measurement of S_z gives one of $n = 2s+1$ outcomes,*

$$S_z = m\hbar, \quad m = -s, -s+1, \dots, +s, \quad (12)$$

so s is found by counting beams. Silver gives 2 beams, so the electron has $s = \frac{1}{2}$. The deuteron (a proton and a neutron with aligned spins) gives 3 beams, $S_z = +\hbar, 0, -\hbar$, so $s = 1$.

Remark 6. S_z is a component. The magnitude is $|\vec{S}| = \hbar\sqrt{s(s+1)}$, which for the electron is $\hbar\sqrt{3}/2$, not $\hbar/2$.

Definition 15 (Ket). *A ket $|\cdot\rangle$ is a label for a quantum state, treated as a vector. For electron spin, the two outcomes of an SGz give*

$$|+\hat{z}\rangle \equiv |\uparrow\rangle \text{ for } S_z = +\frac{\hbar}{2}, \quad |-\hat{z}\rangle \equiv |\downarrow\rangle \text{ for } S_z = -\frac{\hbar}{2},$$

and likewise $|\pm\hat{x}\rangle$ for $S_x = \pm\hbar/2$, and so on for any axis. We follow Townsend and use the $|\pm\hat{n}\rangle$ labels.

3.4 Sequential Stern-Gerlach experiments

We write SG_z for a device whose gradient points along z , and SG_x for the same device rotated by 90° about the beam axis.

Definition 16 (Filter). *Each device sends atoms into two beams. A filter blocks one of them.*

Remark 7. *A beam straight from the oven always splits 50/50 in any SG device, whatever its orientation.*

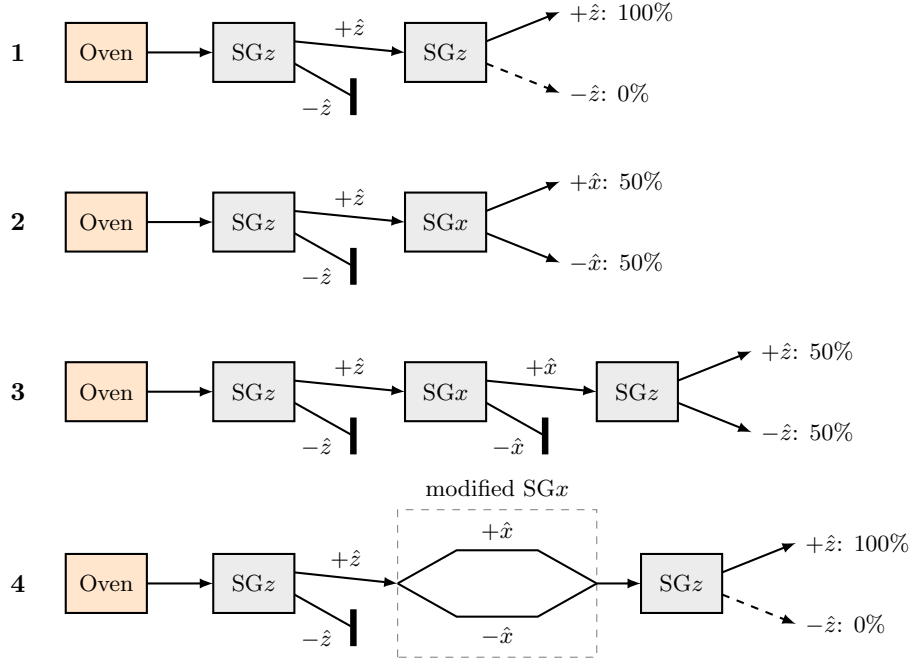


Figure 2: Sequential Stern-Gerlach experiments. Thick bars block a beam. Percentages are the fractions of atoms entering the last device. Every oven starts with N_0 atoms.

Remark 8. *The lecture notes use N_0 for the number of atoms entering the second device rather than leaving the oven, so their counts are twice the ones below.*

3.4.1 Experiment 1: SG_z then SG_z

Filter the $+\hat{z}$ beam and send it into a second SG_z . Every atom comes out $+\hat{z}$. The measurement is repeatable: an atom found with $S_z = +\hbar/2$ gives the same answer again, so it left the first device in $|+\hat{z}\rangle$. We end with $N_0/2$ atoms in $|+\hat{z}\rangle$.

3.4.2 Experiment 2: SG_z then SG_x

Filter $+\hat{z}$ and send it into an SG_x. Half come out $+\hat{x}$, half $-\hat{x}$: knowing S_z tells us nothing about S_x . We end with $N_0/4$ in each of $|\pm\hat{x}\rangle$.

3.4.3 Experiment 3: SG_z, SG_x, then SG_z

Filter $+\hat{z}$, filter $+\hat{x}$, then measure S_z again. The beam splits 50/50 even though every atom was $+\hat{z}$ at the start. Blocking $+\hat{x}$ instead gives the same result. We end with $N_0/8$ in each of $|\pm\hat{z}\rangle$.

Result 5. *No state has definite values of both S_x and S_z . Measuring S_x destroys what we knew about S_z .*

3.4.4 Experiment 4: SG_z, modified SG_x, then SG_z

Definition 17 (Modified SG device). *An SG device that splits the beam according to S_n and then recombines the two beams, so both outcomes exit at the same place without being recorded.*

Filter $+\hat{z}$, pass it through a modified SG_x with neither path blocked, then measure S_z . Every atom comes out $+\hat{z}$: all $N_0/2$ end in $|\hat{z}\rangle$.

Result 6. *The atoms cannot be going half through $+\hat{x}$ and half through $-\hat{x}$, since Experiment 3 showed each half alone gives 50/50. Instead the two paths interfere: as with water or light waves, contributions from the two paths add and can cancel, here cancelling the $-\hat{z}$ outcome. No definite path can be assigned to an atom. Section 4 makes this precise.*

4 Quantum States

September 11-
18, 2026

Throughout, an n -level system is one whose measurements have n possible outcomes. For spin s , $n = 2s + 1$, so spin- $\frac{1}{2}$ has $n = 2$.

4.1 Kets, bras and bases

Definition 18 (Bra and inner product). *Every ket (Def. 15) $|\psi\rangle$ has a matching bra $\langle\psi|$. A bra followed by a ket gives a complex number, the inner product $\langle\phi|\psi\rangle \in \mathbb{C}$.*

Definition 19 (Orthonormal basis). *The outcome states $\{|j\rangle : j = 1, \dots, n\}$ of one measurement form a basis that is orthonormal:*

$$\langle i | j \rangle = \delta_{ij}. \quad (13)$$

For spin- $\frac{1}{2}$ this is $\langle \pm\hat{z} | \pm\hat{z} \rangle = 1$ and $\langle \pm\hat{z} | \mp\hat{z} \rangle = 0$, and the same holds for $\{|\pm\hat{x}\rangle\}$ or any other axis.

Remark 9. The label j counts outcomes, not axes. Choosing to measure S_x gives $|1\rangle = |+\hat{x}\rangle$, $|2\rangle = |-\hat{x}\rangle$; choosing S_z gives $|1\rangle = |+\hat{z}\rangle$, $|2\rangle = |-\hat{z}\rangle$. Either way $n = 2$. A ket may also carry several labels, usually the quantum numbers that pin it down: $|s, m\rangle$ for spin, so $|\pm\hat{z}\rangle = |\frac{1}{2}, \pm\frac{1}{2}\rangle$, and for the deuteron $|1, +1\rangle, |1, 0\rangle, |1, -1\rangle$.

Definition 20 (Superposition). Any state is a linear combination of basis kets,

$$|\psi\rangle = \sum_{j=1}^n c_j |j\rangle, \quad c_j \in \mathbb{C}; \quad \text{spin-}\frac{1}{2}: |\psi\rangle = c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle. \quad (14)$$

Proposition 1 (Coefficients are inner products).

$$c_j = \langle j|\psi\rangle. \quad (15)$$

Proof. $\langle i|\psi\rangle = \sum_j c_j \langle i|j\rangle = \sum_j c_j \delta_{ij} = c_i$. $\square$

Definition 21 (Projector). The outer product (a ket times a bra) $P_j = |j\rangle\langle j|$. It is an operator that keeps only the $|j\rangle$ part of a state,

$$P_j |\psi\rangle = |j\rangle\langle j|\psi\rangle = c_j |j\rangle,$$

and projecting twice changes nothing: $P_j^2 = |j\rangle\langle j||j\rangle\langle j| = P_j$.

Proposition 2 (Completeness relation). Because every state can be expanded in the basis, the projectors sum to the identity:

$$\mathbb{1} = \sum_{j=1}^n |j\rangle\langle j|; \quad \text{spin-}\frac{1}{2}: \mathbb{1} = |+\hat{n}\rangle\langle +\hat{n}| + |-\hat{n}\rangle\langle -\hat{n}| \text{ for any axis } \hat{n}. \quad (16)$$

Remark 10 (Inserting the identity). Writing $\mathbb{1}$ in this form in front of a state reproduces its expansion, $|\psi\rangle = \sum_j |j\rangle\langle j|\psi\rangle$. This trick is used in almost every derivation below.

4.2 Measurement and probability

Definition 22 (Observable). A measurable quantity A . Its possible outcomes a_j each come with a state $|a_j\rangle$, and these states form an orthonormal basis. For spin- $\frac{1}{2}$, A can be S_x , S_y or S_z , with outcomes $\pm\hbar/2$.

Principle 1 (Born rule). The probability that a system in $|\psi\rangle$ is found in $|\phi\rangle$ is

$$P(\phi|\psi) = |\langle\phi|\psi\rangle|^2 \in [0, 1]. \quad (17)$$

In particular $P(A = a_j) = |\langle a_j|\psi\rangle|^2$, e.g. $P(S_z = \pm\hbar/2) = |c_{\pm}|^2$.

Definition 23 (Probability amplitude). The complex number $\langle\phi|\psi\rangle$ whose modulus squared is the probability.

Remark 11. The Born rule has no deeper derivation: it is a postulate that agrees with every experiment. As David Mermin put it, "shut up and calculate."

4.2.1 Bras of superpositions

Write the bra with unknown coefficients, $\langle\psi| = c'_+ \langle+\hat{z}| + c'_- \langle-\hat{z}|$. By orthonormality,

$$\langle\psi|\psi\rangle = c'_+ c_+ + c'_- c_-.$$

A system in $|\psi\rangle$ is found in $|\psi\rangle$ with certainty, and it is always found in one of the two beams, so this must equal $|c_+|^2 + |c_-|^2 = 1$. That forces $c'_\pm = c_\pm^*$.

Result 7. *Going from a ket to its bra, conjugate the coefficients:*

$$\langle\psi| = \sum_j c_j^* \langle j|, \quad \langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*. \quad (18)$$

4.2.2 Normalization

Proposition 3 (Normalization). *$\langle\psi|\psi\rangle = 1$ is the statement that the probabilities of all outcomes add to 1.*

Proof.

$$\langle\psi|\psi\rangle = \sum_{i,j} c_i^* c_j \langle a_i|a_j\rangle = \sum_j |c_j|^2 = \sum_j P(A = a_j) = 1.$$

□

Example. For $|\psi\rangle = \frac{1}{\sqrt{3}}|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}|-\hat{z}\rangle$, $c_+ = 1/\sqrt{3}$ and $c_- = i\sqrt{2/3}$, so $P(S_z = +\hbar/2) = 1/3$ and $P(S_z = -\hbar/2) = |i|^2 \cdot 2/3 = 2/3$. They add to 1, so the state is normalized.

4.2.3 Global and relative phase

Definition 24 (Global and relative phase). *A global phase multiplies the whole ket: $e^{i\alpha}|\psi\rangle$ with $\alpha \in \mathbb{R}$. A relative phase sits between the terms of a superposition, like the $e^{i\delta}$ in $c_+|+\hat{z}\rangle + e^{i\delta}c_-|-\hat{z}\rangle$, and cannot be pulled out front.*

Proposition 4. *A global phase changes no probability: $|\langle\phi|e^{i\alpha}\psi\rangle|^2 = |e^{i\alpha}|^2 |\langle\phi|\psi\rangle|^2 = |\langle\phi|\psi\rangle|^2$ for every $|\phi\rangle$.*

Result 8. *$|\psi\rangle$ and $e^{i\alpha}|\psi\rangle$ are the same physical state, so global phases can be dropped. Relative phases cannot: they leave the S_z probabilities alone but change the S_x and S_y ones (Section 4.4).*

4.3 Expectation value and uncertainty

Definition 25 (Expectation value). *The average outcome, weighted by the probabilities:*

$$\langle A \rangle = \sum_j a_j P(A = a_j) = \sum_j a_j |\langle a_j|\psi\rangle|^2; \quad \langle S_z \rangle = \frac{\hbar}{2}|c_+|^2 - \frac{\hbar}{2}|c_-|^2. \quad (19)$$

Remark 12. For N measurements on identically prepared atoms, with $N_{\pm}$ giving $\pm\hbar/2$, the measured average is $\langle S_z \rangle_N = \frac{\hbar}{2} \frac{N_+ - N_-}{N}$, which tends to $\langle S_z \rangle$ as $N \rightarrow \infty$.

Definition 26 (Uncertainty). The standard deviation (typical spread of outcomes around the average):

$$(\Delta A)^2 = \langle (A - \langle A \rangle)^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2, \quad \langle A^2 \rangle = \sum_j a_j^2 |\langle a_j | \psi \rangle|^2. \quad (20)$$

Remark 13. For spin- $\frac{1}{2}$ both outcomes square to $\hbar^2/4$, so $\langle S_z^2 \rangle = \hbar^2/4$ for every state.

Example. For $|+\hat{x}\rangle$, Experiment 3 gives both S_z probabilities $1/2$, so $\langle S_z \rangle = 0$ and $\Delta S_z = \hbar/2$. The average is zero but every single measurement gives $\pm\hbar/2$: maximal spread.

Example. For $|\psi\rangle = \sqrt{\frac{1}{3}}|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}|-\hat{z}\rangle$,

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{3} - \frac{\hbar}{2} \cdot \frac{2}{3} = -\frac{\hbar}{6}, \quad (\Delta S_z)^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{36} = \frac{2}{9}\hbar^2, \quad \Delta S_z = \frac{\sqrt{2}}{3}\hbar.$$

Remark 14 (Shot noise). The scatter of a finite-sample average around the true expectation value,

$$|\langle S_z \rangle_N - \langle S_z \rangle| \sim \frac{\Delta S_z}{\sqrt{N}}.$$

It comes from the randomness of individual quantum outcomes, not from a faulty apparatus, and is absent only when $\Delta S_z = 0$, i.e. when the state has a definite S_z .

4.4 Spin states along x and y

4.4.1 The state $|+\hat{x}\rangle$

Experiment 3 gives $|\langle \pm\hat{z} | +\hat{x} \rangle|^2 = \frac{1}{2}$, so both amplitudes have modulus $1/\sqrt{2}$ with their own phases:

$$|+\hat{x}\rangle = \frac{e^{i\delta_+}}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle), \quad \delta = \delta_- - \delta_+.$$

Dropping the global phase $e^{i\delta_+}$,

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle).$$

Experiment 3 alone does not fix the relative phase δ .

4.4.2 Experiment 5: measuring S_y

Definition 27 (Cyclic permutation). *Relabelling the axes $x \rightarrow y \rightarrow z \rightarrow x$ (or the reverse). It is a rotation of the coordinate system, so no physics changes.*

Experiment 5 is Experiment 3 cyclically permuted: filter $+\hat{z}$, then measure S_y . By symmetry it also splits 50/50, so every cross-axis overlap has modulus squared $\frac{1}{2}$:

$$|\langle +\hat{x} | +\hat{z} \rangle|^2 = |\langle +\hat{y} | +\hat{z} \rangle|^2 = |\langle -\hat{x} | +\hat{z} \rangle|^2 = \frac{1}{2}.$$

The same argument as for $|+\hat{x}\rangle$ gives, after dropping global phases,

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm e^{i\delta} |-\hat{z}\rangle), \quad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm e^{i\gamma} |-\hat{z}\rangle).$$

4.4.3 Fixing the phases

Using orthonormality of the $\hat{z}$ basis,

$$|\langle +\hat{x} | +\hat{y} \rangle|^2 = \left| \frac{1}{2} (1 + e^{i(\gamma-\delta)}) \right|^2 = \frac{1}{2} (1 + \cos(\gamma - \delta)).$$

Setting this to $\frac{1}{2}$ forces $\gamma - \delta = \pm\pi/2$. The rest is convention.

Result 9 (Phase convention). *With $\delta = 0$ and $\gamma = +\pi/2$,*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm |-\hat{z}\rangle), \quad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm i |-\hat{z}\rangle), \quad (21)$$

$$\langle \pm\hat{x} | = \frac{1}{\sqrt{2}} (\langle +\hat{z} | \pm \langle -\hat{z} |), \quad \langle \pm\hat{y} | = \frac{1}{\sqrt{2}} (\langle +\hat{z} | \mp i \langle -\hat{z} |). \quad (22)$$

Remark 15. *Complex coefficients are required: with real ones the relative phases could only be 0 or π , and $\gamma - \delta = \pm\pi/2$ would be impossible.*

Example. For $|\psi\rangle = \sqrt{\frac{1}{3}} |+\hat{z}\rangle + i\sqrt{\frac{2}{3}} |-\hat{z}\rangle$, compute $\langle S_y \rangle$. Using $\langle \pm\hat{y} | +\hat{z} \rangle = \frac{1}{\sqrt{2}}$ and $\langle \pm\hat{y} | -\hat{z} \rangle = \mp \frac{i}{\sqrt{2}}$,

$$\langle \pm\hat{y} | \psi \rangle = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} \pm \sqrt{\frac{2}{3}} \right), \quad P(S_y = \pm \frac{\hbar}{2}) = \frac{1}{2} \pm \frac{\sqrt{2}}{3},$$

which add to 1. Then

$$\langle S_y \rangle = \frac{\hbar}{2} \left[\left(\frac{1}{2} + \frac{\sqrt{2}}{3} \right) - \left(\frac{1}{2} - \frac{\sqrt{2}}{3} \right) \right] = \frac{\sqrt{2}}{3} \hbar.$$

The factor i did nothing for S_z , but as a relative phase it is exactly what tilts the spin toward $+\hat{y}$.

5 Matrix Mechanics

5.1 States as column vectors

Definition 28 (Representation of a state). *In a basis, a ket is represented by the column of its coefficients and its bra by the conjugate row (the adjoint, $\dagger$). In the S_z basis:*

$$|\psi\rangle \xrightarrow{S_z} \psi = \begin{pmatrix} c_+ \\ c_- \end{pmatrix}, \quad \langle\psi| \xrightarrow{S_z} \psi^\dagger = (c_+^* \quad c_-^*), \quad c_\pm = \langle\pm\hat{z}|\psi\rangle. \quad (23)$$

Remark 16. Write $|\psi\rangle \xrightarrow{S_z} \psi$, not $|\psi\rangle = \psi$: the ket is basis-independent, the column is not.

Example.

$$|\pm\hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}, \quad |\pm\hat{y}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}.$$

Problem 2 (Other bases). *In the S_x basis, $|\psi\rangle \xrightarrow{S_x} \begin{pmatrix} \langle+\hat{x}|\psi\rangle \\ \langle-\hat{x}|\psi\rangle \end{pmatrix}$, and similarly for S_y . Write the column representing $|+\hat{z}\rangle$ in the S_z , S_x and S_y bases.*

Proposition 5 (Inner product as a matrix product). *If $|\phi\rangle \xrightarrow{S_z} \phi$ and $|\psi\rangle \xrightarrow{S_z} \psi$, then $\langle\phi|\psi\rangle = \phi^\dagger\psi$ (row times column), so $P(\phi|\psi) = |\phi^\dagger\psi|^2$ and $\langle\psi|\psi\rangle = \psi^\dagger\psi$.*

Remark 17. *For a state given in another basis, replace each basis ket by its S_z column and add. For $|\psi\rangle = a|+\hat{y}\rangle + b|-\hat{y}\rangle$,*

$$\psi \xrightarrow{S_z} \frac{a}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} + \frac{b}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a+b \\ i(a-b) \end{pmatrix}.$$

Checking $\psi^\dagger\psi = 1$ before going further catches most mistakes. A cleaner systematic method is in Section 5.4.

Problem 3 (A probability by matrix mechanics). *Using matrix mechanics, find $P(S_x = +\hbar/2)$ for the state $|\psi\rangle = \sqrt{\frac{1}{5}}|+\hat{z}\rangle + i\sqrt{\frac{4}{5}}|-\hat{z}\rangle$.*

5.2 Operators as matrices

Definition 29 (Linear operator). *A map A from kets to kets that respects superpositions, $A(a|\psi\rangle + b|\phi\rangle) = aA|\psi\rangle + bA|\phi\rangle$. Every operator in this course is linear.*

Definition 30 (Matrix of an operator). *In a basis $\{|j\rangle\}$, A is represented by the matrix of matrix elements*

$$A_{ij} = \langle i|A|j\rangle. \quad (24)$$

Proposition 6 (Operator acting on a state). $A|\psi\rangle$ is represented by the matrix times the column, $A\psi$.

Proof. The k th entry of the column for $A|\psi\rangle$ is $\langle k|A|\psi\rangle$. Inserting the identity,

$$\langle k|A|\psi\rangle = \sum_j \langle k|A|j\rangle \langle j|\psi\rangle = \sum_j A_{kj} c_j,$$

which is row k of A times the column ψ . $\square$

Example. The projectors and identity in the S_z basis:

$$P_{+\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbb{1} \xrightarrow{S_z} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and indeed $P_{+\hat{z}}\psi = (c_+, 0)^T$.

Definition 31 (Adjoint). The adjoint $A^\dagger$ is the conjugate transpose: $(A^\dagger)_{ij} = A_{ji}^*$. In operator form, writing $A = \sum_{i,j} A_{ij} |i\rangle \langle j|$,

$$A^\dagger = \sum_{i,j} A_{ij}^* |j\rangle \langle i|. \quad (25)$$

For real entries it is just the transpose.

Proposition 7 (Bra of a transformed state). If $|\psi'\rangle = A|\psi\rangle$, then $\langle\psi'| = \langle\psi|A^\dagger$.

Proof. The column of $|\psi'\rangle$ is $A\psi$, and its bra is the conjugate transpose, $(A\psi)^\dagger = \psi^\dagger A^\dagger$. $\square$

Proposition 8 (Adjoint of a product).

$$(AB)^\dagger = B^\dagger A^\dagger. \quad (26)$$

The order reverses; $A^\dagger B^\dagger$ is a different matrix in general.

5.3 Observables as operators

Definition 32 (Eigenstate and eigenvalue). $|a\rangle$ is an eigenstate of A with eigenvalue a if

$$A|a\rangle = a|a\rangle, \quad (27)$$

the familiar $Mv = \lambda v$ in matrix form. A general operator turns a state into a different state; eigenstates come back as themselves times a number. A basis made of eigenstates of A is called an eigenbasis of A .

Proposition 9 (Operator of an observable). For an observable with outcomes a_j and states $|a_j\rangle$, define

$$\hat{A} = \sum_j a_j |a_j\rangle \langle a_j|. \quad (28)$$

Then $\hat{A} |a_j\rangle = a_j |a_j\rangle$ (the outcomes are its eigenvalues, the outcome states its eigenstates), and

$$\langle A \rangle = \langle \psi | \hat{A} | \psi \rangle, \quad \langle A^2 \rangle = \langle \psi | \hat{A}^2 | \psi \rangle. \quad (29)$$

Proof. $\hat{A} |a_k\rangle = \sum_j a_j |a_j\rangle \delta_{jk} = a_k |a_k\rangle$, and $\langle \psi | \hat{A} | \psi \rangle = \sum_j a_j \langle \psi | a_j \rangle \langle a_j | \psi \rangle = \sum_j a_j |\langle a_j | \psi \rangle|^2 = \langle A \rangle$. Likewise $\hat{A}^2 = \sum_j a_j^2 |a_j\rangle \langle a_j|$. $\square$

Proposition 10 (An operator is diagonal in its own eigenbasis).

$$\langle a_i | \hat{A} | a_j \rangle = a_j \delta_{ij}, \quad \hat{A} \xrightarrow{|a_j\rangle} D = \begin{pmatrix} a_1 & & 0 \\ & \ddots & \\ 0 & & a_n \end{pmatrix}. \quad (30)$$

Definition 33 (Spin operators). Applying this to spin, for $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$,

$$\hat{S}_n = \frac{\hbar}{2} (|+\hat{n}\rangle \langle +\hat{n}| - |-\hat{n}\rangle \langle -\hat{n}|), \quad \hat{S}_n |\pm\hat{n}\rangle = \pm \frac{\hbar}{2} |\pm\hat{n}\rangle. \quad (31)$$

In the S_z basis its matrix elements are

$$\langle i | \hat{S}_n | j \rangle = \frac{\hbar}{2} \left(\langle i | +\hat{n} \rangle \langle +\hat{n} | j \rangle - \langle i | -\hat{n} \rangle \langle -\hat{n} | j \rangle \right), \quad i, j \in \{+\hat{z}, -\hat{z}\}.$$

Problem 4 (The matrix of $\hat{S}_z$). Find the matrix representation of $\hat{S}_z$ in the S_z basis.

Problem 5 (The matrix of $\hat{S}_x$). Find the matrix representation of $\hat{S}_x$ in the S_z basis.

Problem 6 (The matrix of $\hat{S}_y$). Find the matrix representation of $\hat{S}_y$ in the S_z basis.

Remark 18 (Shortcut). Write each projector as a column times its conjugate row, e.g.

$$|\pm\hat{x}\rangle \langle \pm\hat{x}| \xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix} (1 \quad \pm 1) = \frac{1}{2} \begin{pmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{pmatrix},$$

then subtract and multiply by $\hbar/2$.

Definition 34 (Pauli matrices). The results, all in the S_z basis:

$$\hat{S}_j = \frac{\hbar}{2} \sigma_j, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (32)$$

Each equals its own adjoint and squares to $\mathbb{1}$, so $\hat{S}_j^2 = \frac{\hbar^2}{4} \mathbb{1}$.

Problem 7 (Expectation value and uncertainty by matrix mechanics). For $|\psi\rangle = \sqrt{\frac{1}{5}} |+\hat{y}\rangle + i\sqrt{\frac{4}{5}} |-\hat{y}\rangle$, find $\langle S_z \rangle$ and ΔS_z using matrix mechanics.

5.4 Change of basis

Take an n -level system with two orthonormal bases, $\{|j\rangle\}$ and $\{|\alpha\rangle\}$, $j, \alpha = 1, \dots, n$. Within each basis the overlaps are Kronecker deltas, but across bases they are not: $\langle +\hat{x} | +\hat{z} \rangle = \frac{1}{\sqrt{2}}$ is neither 0 nor 1. The same ket then has two columns,

$$|\psi\rangle \xrightarrow{|j\rangle} \psi^j = \begin{pmatrix} \langle 1|\psi\rangle \\ \vdots \\ \langle n|\psi\rangle \end{pmatrix}, \quad |\psi\rangle \xrightarrow{|\alpha\rangle} \psi^\alpha = \begin{pmatrix} \langle \alpha=1|\psi\rangle \\ \vdots \\ \langle \alpha=n|\psi\rangle \end{pmatrix}.$$

Inserting $\mathbb{1} = \sum_\alpha |\alpha\rangle \langle \alpha|$ into each $|j\rangle$ amplitude,

$$\langle j|\psi\rangle = \sum_\alpha \langle j|\alpha\rangle \langle \alpha|\psi\rangle.$$

Result 10 (Change-of-basis matrix).

$$\psi^j = U\psi^\alpha, \quad \psi^\alpha = U^\dagger \psi^j, \quad U_{j\alpha} = \langle j|\alpha\rangle. \quad (33)$$

The columns of U are the new basis kets $|\alpha\rangle$ written in the old basis $\{|j\rangle\}$.

Definition 35 (Unitary operator). U is unitary if $U^\dagger U = U U^\dagger = \mathbb{1}$, i.e. its inverse is its adjoint.

Theorem 1. The change-of-basis matrix U is unitary.

Proof. With $(U^\dagger)_{\alpha j} = \langle j|\alpha\rangle^* = \langle \alpha|j\rangle$ and completeness of $\{|j\rangle\}$,

$$(U^\dagger U)_{\alpha\beta} = \sum_j \langle \alpha|j\rangle \langle j|\beta\rangle = \langle \alpha|\beta\rangle = \delta_{\alpha\beta}.$$

Completeness of $\{|\alpha\rangle\}$ gives $U U^\dagger = \mathbb{1}$ the same way. $\square$

Remark 19. So normalization does not depend on the basis: $\psi^{j\dagger} \psi^j = \psi^{\alpha\dagger} U^\dagger U \psi^\alpha = \psi^{\alpha\dagger} \psi^\alpha$.

Example (From S_x to S_z). With $|j\rangle = |\pm\hat{z}\rangle$ and $|\alpha\rangle = |\pm\hat{x}\rangle$, the columns of U are the S_z columns of $|\pm\hat{x}\rangle$:

$$U = \begin{pmatrix} \langle +\hat{z} | +\hat{x} \rangle & \langle +\hat{z} | -\hat{x} \rangle \\ \langle -\hat{z} | +\hat{x} \rangle & \langle -\hat{z} | -\hat{x} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \psi^z = U\psi^x. \quad (34)$$

It is real and symmetric, so $U^\dagger = U$. The matrix taking the S_y basis to S_z has columns $\frac{1}{\sqrt{2}}(1, \pm i)^T$.

Problem 8 (Expectation value from the S_x basis). For $|\psi\rangle = \frac{1}{\sqrt{3}}|+\hat{x}\rangle + i\sqrt{\frac{2}{3}}|-\hat{x}\rangle$, find $\langle S_z \rangle = \psi^{z\dagger} \hat{S}_z \psi^z$ using matrix mechanics.

5.4.1 Operators in another basis

Let $\{|j\rangle\} = \{|a_j\rangle\}$ be the eigenbasis of $\hat{A}$, so $\hat{A} \xrightarrow{|a_j\rangle} D$ (diagonal), and $U_{j\alpha} = \langle a_j | \alpha \rangle$.

Proposition 11 (Similarity transformation).

$$\hat{A} \xrightarrow{|\alpha\rangle} A^\alpha = U^\dagger D U. \quad (35)$$

Rewriting the same operator in a new basis this way is called a similarity transformation.

Proof. Insert the identity twice:

$$\langle \alpha | \hat{A} | \beta \rangle = \sum_{i,j} \langle \alpha | a_i \rangle \langle a_i | \hat{A} | a_j \rangle \langle a_j | \beta \rangle = \sum_{i,j} (U^\dagger)_{\alpha i} D_{ij} U_{j\beta} = (U^\dagger D U)_{\alpha\beta}.$$

□

Result 11 (Expectation value in any basis).

$$\langle A \rangle = \psi^{j\dagger} D \psi^j = \psi^{\alpha\dagger} [U^\dagger D U] \psi^\alpha. \quad (36)$$

Either move the state to the eigenbasis, or move the operator to the state's basis. Both give the same number.

Problem 9 ($\hat{S}_z$ in the S_x basis). Using U from Eq. (34), find the matrix of $\hat{S}_z$ in the S_x basis, and use it to redo the problem “Expectation value from the S_x basis” without changing the state's basis.

6 Rotations

6.1 The rotation operator

Definition 36 (Rotation operator). $R(\theta\hat{n})$ is the linear operator that rotates a state by angle θ about the axis $\hat{n}$: $|\psi'\rangle = R(\theta\hat{n})|\psi\rangle$.

Example. $R(\frac{\pi}{2}\hat{y})$ turns $\hat{z}$ into $\hat{x}$, so it takes $|\pm\hat{z}\rangle$ to $|\pm\hat{x}\rangle$ up to phase, and by linearity

$$R\left(\frac{\pi}{2}\hat{y}\right)(c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle) = c_+|+\hat{x}\rangle + c_-|-\hat{x}\rangle.$$

Up to a phase on one column, its S_z matrix is the change-of-basis matrix U of Eq. (34) (exact form in Section 6.4).

Remark 20 (Why the bra carries $R^\dagger$). By the bra-of-a-transformed-state proposition, $\langle\psi'| = \langle\psi|R^\dagger$. Using $\langle\psi|R$ instead fails: with $|+\hat{x}\rangle = R(\frac{\pi}{2}\hat{y})|+\hat{z}\rangle$ it would give

$$\langle+\hat{x}|+\hat{x}\rangle = \langle+\hat{z}|R\left(\frac{\pi}{2}\hat{y}\right)R\left(\frac{\pi}{2}\hat{y}\right)|+\hat{z}\rangle = \langle+\hat{z}|R(\pi\hat{y})|+\hat{z}\rangle \propto \langle+\hat{z}|-\hat{z}\rangle = 0,$$

since a π rotation about $\hat{y}$ flips $+\hat{z}$ to $-\hat{z}$. That contradicts $\langle+\hat{x}|+\hat{x}\rangle = 1$.

Theorem 2. *Rotation operators are unitary: $R^\dagger R = RR^\dagger = \mathbb{1}$.*

Proof. A rotation must preserve normalization: $\langle \psi' | \psi' \rangle = \langle \psi | R^\dagger R | \psi \rangle = 1$ for every $|\psi\rangle$, which forces $R^\dagger R = \mathbb{1}$. $\square$

Remark 21. *If a unitary U maps $|\psi\rangle$ to $c|\phi\rangle$ with $|\phi\rangle$ normalized, then $1 = \langle \psi | U^\dagger U | \psi \rangle = |c|^2$, so $c = e^{i\varphi}$. In particular, eigenvalues of a unitary are pure phases, and U leaves a physical state unchanged only when that state is an eigenstate of U .*

6.2 Generators of rotations

Rotations about the same axis add, $R(\theta_1 \hat{z})R(\theta_2 \hat{z}) = R((\theta_1 + \theta_2) \hat{z})$, so a finite rotation is $N \gg 1$ small ones,

$$R(\theta \hat{z}) = \left[R\left(\frac{\theta}{N} \hat{z}\right) \right]^N. \quad (37)$$

Taylor expanding one small step about the identity and naming the first-order operator $-iJ_z/\hbar$,

$$R\left(\frac{\theta}{N} \hat{z}\right) = \mathbb{1} - \frac{i}{\hbar} J_z \frac{\theta}{N} + \mathcal{O}\left(\frac{1}{N^2}\right). \quad (38)$$

Result 12. *By the problem “The exponential as a limit”,*

$$R(\theta \hat{z}) = \lim_{N \rightarrow \infty} \left(\mathbb{1} - \frac{i}{\hbar} J_z \frac{\theta}{N} \right)^N = e^{-iJ_z \theta / \hbar}, \quad (39)$$

and likewise $R(\theta \hat{n}) = e^{-iJ_n \theta / \hbar}$ for any axis.

Definition 37 (Generator of rotations). *The operator J_z above: the thing that, exponentiated, produces rotations about $\hat{z}$. It has units of $\hbar$, i.e. of angular momentum ($[\hbar] = [\vec{r} \times \vec{p}]$), so $\vec{J}$ is identified as the angular momentum operator. For spin- $\frac{1}{2}$, $\vec{J} = \vec{S}$ and $R(\theta \hat{n}) = e^{-i\theta \hat{S}_n / \hbar}$.*

Definition 38 (Hermitian operator). *A is Hermitian if it equals its own adjoint, $A = A^\dagger$ (e.g. the Pauli matrices).*

Proposition 12. *The generator J_z is Hermitian.*

Proof. To first order in θ , $R = \mathbb{1} - i\theta J_z / \hbar$ and $R^\dagger = \mathbb{1} + i\theta J_z^\dagger / \hbar$. Unitarity gives

$$\mathbb{1} = R^\dagger R = \mathbb{1} + \frac{i\theta}{\hbar} (J_z^\dagger - J_z) + \mathcal{O}(\theta^2),$$

so $J_z^\dagger = J_z$. $\square$

Principle 2. *Every observable is represented by a Hermitian operator.*

Proposition 13. *The eigenvalues of a Hermitian operator are real, so measurement outcomes are real.*

Proof. If $A|a\rangle = a|a\rangle$, the bra is $\langle a|A^\dagger = a^* \langle a|$, so $a \langle a|a\rangle = \langle a|A|a\rangle = \langle a|A^\dagger|a\rangle = a^* \langle a|a\rangle$, and $a = a^*$. $\square$

6.3 Functions of an operator

Proposition 14 (Function of an operator on an eigenstate). *If $\hat{A}|a\rangle = a|a\rangle$ and f has a Taylor series, then*

$$f(\hat{A})|a\rangle = f(a)|a\rangle. \quad (40)$$

Proof. $f(\hat{A})$ is defined by its Taylor series, and $\hat{A}^n|a\rangle = a^n|a\rangle$, so term by term

$$f(\hat{A})|a\rangle = \sum_n \frac{f^{(n)}(0)}{n!} \hat{A}^n|a\rangle = \sum_n \frac{f^{(n)}(0)}{n!} a^n|a\rangle = f(a)|a\rangle.$$

□

Proposition 15 (Spectral form). *Writing an operator as a sum of its eigenvalues times the projectors onto its eigenstates is its spectral form. Inserting $\mathbb{1} = \sum_j |a_j\rangle\langle a_j|$ and using Proposition 14,*

$$f(\hat{A}) = \sum_j f(a_j) |a_j\rangle\langle a_j|. \quad (41)$$

With $f(x) = x$ this is the observable operator $\hat{A} = \sum_j a_j |a_j\rangle\langle a_j|$.

6.3.1 Rotations about $\hat{z}$

Result 13. *With $f(x) = e^{-i\phi x/\hbar}$ and $\hat{S}_z|\pm\hat{z}\rangle = \pm\frac{\hbar}{2}|\pm\hat{z}\rangle$,*

$$R(\phi\hat{z})|\pm\hat{z}\rangle = e^{\mp i\phi/2}|\pm\hat{z}\rangle, \quad R(\phi\hat{z}) \xrightarrow{S_z} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix}. \quad (42)$$

This makes sense: a rotation about $\hat{z}$ cannot change S_z , so it can only attach phases to the S_z states. In other bases the matrix is not diagonal.

Remark 22. *A full turn, $\phi = 2\pi$, gives $R = -\mathbb{1}$: a spin- $\frac{1}{2}$ state needs 4π to return to itself. The sign is a global phase, so it is invisible on its own.*

Problem 10 ($R(\phi\hat{z})$ in the S_x basis). *Find the matrix representation of $R(\phi\hat{z})$ in the S_x basis.*

6.4 Rotations about any axis

Proposition 16 (Spectral form of a rotation). *For $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$,*

$$R(\phi\hat{n}) = e^{-i\phi/2}|+\hat{n}\rangle\langle+\hat{n}| + e^{i\phi/2}|-\hat{n}\rangle\langle-\hat{n}|. \quad (43)$$

Proof. Spectral form with $f(x) = e^{-i\phi x/\hbar}$, $\hat{A} = \hat{S}_n$, eigenvalues $\pm\hbar/2$. □

Remark 23 (Recipe). *In any basis $\{|i\rangle\}$ this reads $VDV^\dagger$, with $D = \text{diag}(e^{-i\phi/2}, e^{i\phi/2})$ and V the matrix whose columns are $|\pm\hat{n}\rangle$ written in $\{|i\rangle\}$. It is the similarity transformation with the roles of the two bases swapped.*

Proposition 17 ($R(\phi\hat{x})$ in the S_z basis).

$$R(\phi\hat{x}) \xrightarrow{S_z} \begin{pmatrix} \cos \frac{\phi}{2} & -i \sin \frac{\phi}{2} \\ -i \sin \frac{\phi}{2} & \cos \frac{\phi}{2} \end{pmatrix}. \quad (44)$$

Proof. Here $V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = V^\dagger$, so

$$R(\phi\hat{x}) \xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} e^{-i\phi/2} + e^{i\phi/2} & e^{-i\phi/2} - e^{i\phi/2} \\ e^{-i\phi/2} - e^{i\phi/2} & e^{-i\phi/2} + e^{i\phi/2} \end{pmatrix},$$

and Euler's formula gives $2 \cos \frac{\phi}{2}$ and $-2i \sin \frac{\phi}{2}$. $\square$

Remark 24. *This is the same matrix as $R(\phi\hat{z})$ in the S_x basis.*

Proposition 18 ($R(\phi\hat{y})$ in the S_z basis).

$$R(\phi\hat{y}) \xrightarrow{S_z} \begin{pmatrix} \cos \frac{\phi}{2} & -\sin \frac{\phi}{2} \\ \sin \frac{\phi}{2} & \cos \frac{\phi}{2} \end{pmatrix}. \quad (45)$$

Proof. Now $V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$ and $V^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \neq V$. Then

$$R(\phi\hat{y}) \xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} e^{-i\phi/2} + e^{i\phi/2} & -i(e^{-i\phi/2} - e^{i\phi/2}) \\ i(e^{-i\phi/2} - e^{i\phi/2}) & e^{-i\phi/2} + e^{i\phi/2} \end{pmatrix},$$

and with $e^{-i\phi/2} - e^{i\phi/2} = -2i \sin \frac{\phi}{2}$ the off-diagonal entries are $\mp \sin \frac{\phi}{2}$. $\square$

Remark 25. *This matrix is real, so $R(\phi\hat{y})$ never introduces factors of i into an S_z column. At $\phi = \pi$ it gives $R(\pi\hat{y})|+\hat{z}\rangle = |-\hat{z}\rangle$, the flip used above to show bras carry $R^\dagger$.*

Remark 26 (Phase caveat). *At $\phi = \pi/2$,*

$$R\left(\frac{\pi}{2}\hat{y}\right) \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix},$$

whose second column is $-|\hat{x}\rangle$, while U in Eq. (34) has $+|\hat{x}\rangle$. They differ only by the phase of one basis ket, which is a free choice, so both are valid.

7 Angular Momentum

7.1 Rotations about different axes

Definition 39 (Commutator).

$$[A, B] = AB - BA. \quad (46)$$

A and B commute if $[A, B] = 0$, i.e. the order in which they act does not matter.

Rotations about different axes do not commute: $R(\theta\hat{x})R(\theta\hat{y}) \neq R(\theta\hat{y})R(\theta\hat{x})$.

Remark 27. Rotate a textbook by 90° about $\hat{x}$ then about $\hat{y}$ (right-hand rule). Repeat from the same start with $\hat{y}$ first. The spine ends up pointing in different directions.

Result 14 (Angular momentum commutation relations).

$$[J_x, J_y] = i\hbar J_z, \quad [J_y, J_z] = i\hbar J_x, \quad [J_z, J_x] = i\hbar J_y, \quad (47)$$

or compactly $\vec{J} \times \vec{J} = i\hbar \vec{J}$, with $\vec{J} = J_x\hat{x} + J_y\hat{y} + J_z\hat{z}$.

7.2 Compatible observables

Definition 40 (Degenerate eigenvalue). An eigenvalue is degenerate if more than one independent eigenstate has it. Those eigenstates span a subspace: a smaller vector space inside the full one, closed under linear combinations.

Theorem 3. If $[A, B] = 0$, then A and B have a common eigenbasis.

Proof. Let $A|a\rangle = a|a\rangle$. Since $AB = BA$,

$$A(B|a\rangle) = BA|a\rangle = a(B|a\rangle),$$

so $B|a\rangle$ is also an eigenstate of A with eigenvalue a .

Nondegenerate case. If $|a\rangle$ is the only state with eigenvalue a , then $B|a\rangle = b|a\rangle$, so $|a\rangle$ is an eigenstate of both.

Degenerate case. If n states $|a, i\rangle$ share eigenvalue a , then B maps their subspace into itself, so there it is an $n \times n$ matrix $B_{ij} = \langle a, i|B|a, j\rangle$. Diagonalizing it (finding the basis in which it is diagonal) gives eigenstates

$$|a, b_j\rangle = \sum_i |a, i\rangle \langle a, i|a, b_j\rangle, \quad A|a, b_j\rangle = a|a, b_j\rangle, \quad B|a, b_j\rangle = b_j|a, b_j\rangle,$$

which are still eigenstates of A because they are combinations of states with the same eigenvalue a . $\square$

Definition 41 (Compatible observables). Observables that commute, and so can have definite values at the same time. Otherwise they are incompatible.

Remark 28. $[J_x, J_y] = i\hbar J_z \neq 0$, so the components of angular momentum are incompatible. This is the operator version of Experiment 3.

7.3 Total angular momentum

Definition 42 (Square of the angular momentum).

$$\vec{J}^2 = J_x^2 + J_y^2 + J_z^2. \quad (48)$$

Proposition 19. $\vec{J}^2$ commutes with every component of $\vec{J}$; in particular $[\vec{J}^2, J_z] = 0$.

Proof. Using $[A, BC] = B[A, C] + [A, B]C$:

$$\begin{aligned} [J_z, J_x^2] &= J_x[J_z, J_x] + [J_z, J_x]J_x = i\hbar(J_xJ_y + J_yJ_x), \\ [J_z, J_y^2] &= J_y[J_z, J_y] + [J_z, J_y]J_y = -i\hbar(J_yJ_x + J_xJ_y), \\ [J_z, J_z^2] &= 0, \end{aligned}$$

using $[J_z, J_y] = -i\hbar J_x$. The first two cancel, so $[J_z, \vec{J}^2] = 0$. The same steps work for x and y . $\square$

Remark 29. $\vec{J}^2$ and J_z are compatible, so they share an eigenbasis.

A Solutions

Solution 1 (The exponential as a limit). *Take the logarithm:*

$$\ln \left(1 + \frac{x}{N}\right)^N = N \ln \left(1 + \frac{x}{N}\right).$$

With $\ln(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \dots$ and $u = x/N$,

$$N \ln \left(1 + \frac{x}{N}\right) = x - \frac{x^2}{2N} + \frac{x^3}{3N^2} - \dots.$$

Every term after the first vanishes as $N \rightarrow \infty$, so the logarithm tends to x and

$$\lim_{N \rightarrow \infty} \left(1 + \frac{x}{N}\right)^N = e^x.$$

Solution 2 (Other bases).

$$\begin{aligned} |+\hat{z}\rangle &\xrightarrow{S_z} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \\ |+\hat{z}\rangle &\xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x} | +\hat{z} \rangle \\ \langle -\hat{x} | +\hat{z} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \\ |+\hat{z}\rangle &\xrightarrow{S_y} \begin{pmatrix} \langle +\hat{y} | +\hat{z} \rangle \\ \langle -\hat{y} | +\hat{z} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}. \end{aligned}$$

Solution 3 (A probability by matrix mechanics).

$$\begin{aligned} \psi &\xrightarrow{S_z} \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2i \end{pmatrix}, \quad \langle +\hat{x} | \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix}, \\ \langle +\hat{x} | \psi \rangle &= \frac{1 + 2i}{\sqrt{10}}, \quad P(S_x = +\frac{\hbar}{2}) = \frac{|1 + 2i|^2}{10} = \frac{1}{2}. \end{aligned}$$

Take the modulus of the whole sum, not of each term separately.

Solution 4 (The matrix of $\hat{S}_z$). Every overlap is 0 or 1, so only the diagonal survives:

$$\langle \pm \hat{z} | \hat{S}_z | \pm \hat{z} \rangle = \pm \frac{\hbar}{2}, \quad \hat{S}_z \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

An operator is diagonal in its own eigenbasis, with its outcomes on the diagonal.

Solution 5 (The matrix of $\hat{S}_x$). The overlaps are $\langle +\hat{z} | \pm \hat{x} \rangle = \langle -\hat{z} | \pm \hat{x} \rangle = \frac{1}{\sqrt{2}}$ and $\langle -\hat{z} | -\hat{x} \rangle = -\frac{1}{\sqrt{2}}$, all real. Then

$$\begin{aligned} \langle +\hat{z} | \hat{S}_x | +\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{2} - \frac{1}{2} \right) = 0, & \langle +\hat{z} | \hat{S}_x | -\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{2} + \frac{1}{2} \right) = \frac{\hbar}{2}, \\ \langle -\hat{z} | \hat{S}_x | +\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{2} + \frac{1}{2} \right) = \frac{\hbar}{2}, & \langle -\hat{z} | \hat{S}_x | -\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{2} - \frac{1}{2} \right) = 0, \end{aligned}$$

where the + signs come from the minus in $\langle -\hat{z} | -\hat{x} \rangle$ meeting the minus in front of the second projector. So

$$\hat{S}_x \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Solution 6 (The matrix of $\hat{S}_y$). The overlaps are complex and the bra side is conjugated: $\langle -\hat{z} | \pm \hat{y} \rangle = \pm \frac{i}{\sqrt{2}}$, $\langle \pm \hat{y} | -\hat{z} \rangle = \mp \frac{i}{\sqrt{2}}$, and $\langle +\hat{z} | \pm \hat{y} \rangle = \langle \pm \hat{y} | +\hat{z} \rangle = \frac{1}{\sqrt{2}}$. Then

$$\begin{aligned} \langle +\hat{z} | \hat{S}_y | -\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{\sqrt{2}} \cdot \frac{-i}{\sqrt{2}} - \frac{1}{\sqrt{2}} \cdot \frac{i}{\sqrt{2}} \right) = -\frac{i\hbar}{2}, \\ \langle -\hat{z} | \hat{S}_y | +\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{i}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - \frac{-i}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \right) = +\frac{i\hbar}{2}, \end{aligned}$$

and the diagonal entries vanish because $i \cdot (-i) = 1$ in both terms. So

$$\hat{S}_y \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Solution 7 (Expectation value and uncertainty by matrix mechanics). In the S_z basis,

$$\psi \xrightarrow{S_z} \frac{1}{\sqrt{10}} \left[\begin{pmatrix} 1 \\ i \end{pmatrix} + \begin{pmatrix} 2i \\ 2 \end{pmatrix} \right] = \frac{1}{\sqrt{10}} \begin{pmatrix} 1+2i \\ 2+i \end{pmatrix}.$$

Then

$$\langle S_z \rangle = \psi^\dagger \hat{S}_z \psi = \frac{\hbar}{2} \cdot \frac{1}{10} (1-2i \quad 2-i) \begin{pmatrix} 1+2i \\ -(2+i) \end{pmatrix} = \frac{\hbar}{2} \cdot \frac{5-5}{10} = 0.$$

Since $\hat{S}_z^2 = \frac{\hbar^2}{4} \mathbb{1}$, $\langle S_z^2 \rangle = \frac{\hbar^2}{4}$, so $\Delta S_z = \hbar/2$: both outcomes have probability $\frac{1}{2}$ and the spread is maximal.

Solution 8 (Expectation value from the S_x basis).

$$|\psi\rangle \xrightarrow{S_x} \psi^x = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ i\sqrt{2} \end{pmatrix}.$$

Change to the S_z basis with U from Eq. (34):

$$\psi^z = U\psi^x = \frac{1}{\sqrt{6}} \begin{pmatrix} 1+i\sqrt{2} \\ 1-i\sqrt{2} \end{pmatrix}, \quad \psi^{z\dagger} = \frac{1}{\sqrt{6}} (1-i\sqrt{2} \quad 1+i\sqrt{2}).$$

Then

$$\langle S_z \rangle = \psi^{z\dagger} \hat{S}_z \psi^z = \frac{\hbar}{2} \cdot \frac{1}{6} (|1+i\sqrt{2}|^2 - |1-i\sqrt{2}|^2) = \frac{\hbar}{2} \left(\frac{3}{6} - \frac{3}{6} \right) = 0.$$

Both components have equal modulus, so both S_z outcomes have probability $\frac{1}{2}$.

Solution 9 ($\hat{S}_z$ in the S_x basis). U from Eq. (34) is real and symmetric, so $U^\dagger = U$ and

$$\hat{S}_z \xrightarrow{S_x} U^\dagger \hat{S}_z U = \frac{\hbar}{2} \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This is $\frac{\hbar}{2}\sigma_x$: in the S_x basis, $\hat{S}_z$ looks the way $\hat{S}_x$ looks in the S_z basis.

Keeping the state in the S_x basis,

$$\langle S_z \rangle = \psi^{x\dagger} \hat{S}_z \psi^x = \frac{\hbar}{2} \cdot \frac{1}{3} (1 \quad -i\sqrt{2}) \begin{pmatrix} i\sqrt{2} \\ 1 \end{pmatrix} = \frac{\hbar}{2} \cdot \frac{i\sqrt{2} - i\sqrt{2}}{3} = 0,$$

matching the previous solution. In general, for $\psi^x = (c_+, c_-)^T$ this gives $\langle S_z \rangle = \hbar \operatorname{Re}(c_+^* c_-)$, which vanishes here because the relative phase is i .

Solution 10 ($R(\phi\hat{z})$ in the S_x basis). Expanding the S_x states in $\hat{z}$,

$$R|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} (e^{-i\phi/2} |+\hat{z}\rangle \pm e^{i\phi/2} |-\hat{z}\rangle).$$

Hitting these with $\langle\pm\hat{x}| = \frac{1}{\sqrt{2}} (\langle+\hat{z}| \pm \langle-\hat{z}|)$ and using Euler's formula,

$$\begin{aligned} \langle+\hat{x}| R |+\hat{x}\rangle &= \langle-\hat{x}| R |-\hat{x}\rangle = \frac{1}{2} (e^{-i\phi/2} + e^{i\phi/2}) = \cos \frac{\phi}{2}, \\ \langle+\hat{x}| R |-\hat{x}\rangle &= \langle-\hat{x}| R |+\hat{x}\rangle = \frac{1}{2} (e^{-i\phi/2} - e^{i\phi/2}) = -i \sin \frac{\phi}{2}, \end{aligned}$$

so

$$R(\phi\hat{z}) \xrightarrow{S_x} \begin{pmatrix} \cos \frac{\phi}{2} & -i \sin \frac{\phi}{2} \\ -i \sin \frac{\phi}{2} & \cos \frac{\phi}{2} \end{pmatrix}.$$

Check: at $\phi = \pi/2$, $R|+\hat{x}\rangle \xrightarrow{S_x} \frac{1}{\sqrt{2}}(1, -i)^T$, which is $|+\hat{y}\rangle$ up to a global phase, as rotating $\hat{x}$ by 90° about $\hat{z}$ should give.

B Useful Links

Stern-Gerlach experiment:

<https://www.youtube.com/shorts/QG7kJXxN1Kk>